
Software Requirements Specification

for

BarberHub

Version 1.0 approved

**Prepared by Group 17
Nathanael Setiorahardjo, Gavinn Aloys, Kelvin Leandi**

BINUS University

25 February 2026

Table of Contents

1. Introduction	1
1.1 Purpose	1
1.2 Document Conventions	1
1.3 Intended Audience and Reading Suggestions	1
1.4 Product Scope	2
1.5 References	3
2. Overall Description	4
2.1 Product Perspective	4
2.2 Product Functions	4
2.3 User Classes and Characteristics	4
2.4 Operating Environment	5
2.5 Design and Implementation Constraints	5
2.6 User Documentation	6
2.7 Assumptions and Dependencies	6
3. External Interface Requirements	7
3.1 User Interfaces	7
3.2 Hardware Interfaces	8
3.3 Software Interfaces	8
3.4 Communications Interfaces	8
4. System Features	10
4.1 User Authentication & Registration	10
4.2 Barber Dashboard	11
4.3 Appointment Booking System	11
4.4 Life & Punctuality Enforcement System	12
4.5 Rating & Review System	13
4.6 Booking History & Status Tracking	14
4.7 Admin Management Panel	15
5. Other Nonfunctional Requirements	16
5.1 Performance Requirements	16
5.2 Safety Requirements	16
5.3 Security Requirements	16
5.4 Software Quality Attributes	17
5.5 Business Rules	17
6. Other Requirements	18
6.1 Database Requirements	18
6.2 Browser & Device Support	18
7. Unified Modeling Languages	20
7.1 Use Case Diagram	20
7.2 Activity Diagrams	21

7.3 BarberHub ERD**28****7.4 BarberHub Sequence Diagrams****29**

Revision History

Name	Date	Reason For Changes	Version
Group 17	25/02/26	Progress Chapter 1 content according to team discussion	1.0
Group 17	17/03/26	Progress Chapter 2-3 content according to team discussion	1.1
Group 17	26/03/26	Progress Chapter 4-6 content according to team discussion	1.2
Group 17	02/04/26	Added technical edge cases, database concurrency locking, automated reconciliation (cron jobs), and performance constraints based on engineering architectural review.	1.3
Group 17	13/04/26	Added UML to visualize business logic and application flow.	1.4
Group 17	15/04/26	Added Link towards Figma Prototype to visualize frontend of BarberHub	1.5

1. Introduction

1.1 Purpose

BarberHub is developed as an application that provides an online platform for booking haircut services, integrated with a barber schedule management system and a customer discipline mechanism based on a "life" system. The system aims to improve the operational efficiency of barbershops while delivering a more structured, transparent, and fair reservation experience for both customers and barbers.

Through this application, customers can view real-time barber availability, choose a barber based on preferences such as rating, service price, and years of work experience, and make bookings on available time slots.

BarberHub enforces a "3-life" system for every customer. If a customer is late by more than 20 minutes from their scheduled booking time, one life is deducted. If all lives are exhausted, the customer's account is automatically blocked by the system. This mechanism is designed to minimize no-shows and improve time discipline.

This Software Requirements Specification (SRS) defines the system scope, key features, and functional and non-functional requirements of BarberHub. The document serves as the official reference throughout requirement analysis, system design, implementation, and testing.

1.2 Document Conventions

High-level functioning requirements should always be identified with a unique ID, and it is assumed that any priority assigned to a parent requirement is automatically inherited by its detailed sub-requirements. Within the text, all User Interface Elements (such as buttons or menu items) must be bolded, while User Roles such as Admin or Customer, should be italicized. These conventions are designed to ensure that all developers and testers can quickly scan the document for actionable technical details without misinterpretation in the hierarchy of needs.

1.3 Intended Audience and Reading Suggestions

This SRS document for the BarberHub system is created for all parties involved in the development and operation of the application.

Target Readers:

- **Barbershop Owners:** Parties who use the application to manage operations and barber schedules.
- **Project Manager:** Parties who oversee and manage the application development process.

- **Software Developers:** Programmers who build the system from the logic, database, and feature implementation side.
- **Quality Assurance:** Parties who ensure all features run according to specified requirements.

Reading Guide:

- **For Barbershop Owners and Project Manager:** It is recommended to read Chapters 1 and 2 to understand the overall system overview, business objectives, and application limitations.
- **For Software Developers:** It is recommended to read the entire document, with the main focus on Chapter 3 (functional requirements and system flow) and Chapter 4 (non-functional requirements such as performance and security).
- **For Quality Assurance:** It is recommended to read Chapter 2 to understand the operational context, then Chapters 3 and 4 as the basis for creating test scenarios (test cases).

Stakeholder	Role
Project Manager (Lecturer)	Acts as the project manager and evaluator who reviews, validates, and approves system requirements and overall project direction.
Development Team (Group 17)	Responsible for designing, implementing, and integrating the BarberHub prototype according to the specified requirements.

1.4 Product Scope

BarberHub is developed as a web-based barbershop reservation application that integrates a barber schedule management system with a customer discipline control mechanism. The system is designed to reduce manual queues, increase scheduling certainty, and minimize losses caused by customer no-shows.

The product scope encompasses the management of barber schedules based on time availability as set by the admin or the barbers themselves. The system will display available time slots on the customer dashboard in real-time.

Each barber has a profile that includes:

- Rating (based on customer reviews)
- Service price
- Work experience (in years)

Customers can choose a barber according to their preferences and make bookings on available schedules. The system will record booking time and customer arrival time. If a customer is late by more than 20 minutes from the scheduled time, the system will automatically deduct one life from the customer's total of three lives.

If all lives are exhausted, the customer's account will be blocked and they will be unable to make bookings until administrative action is taken by an admin.

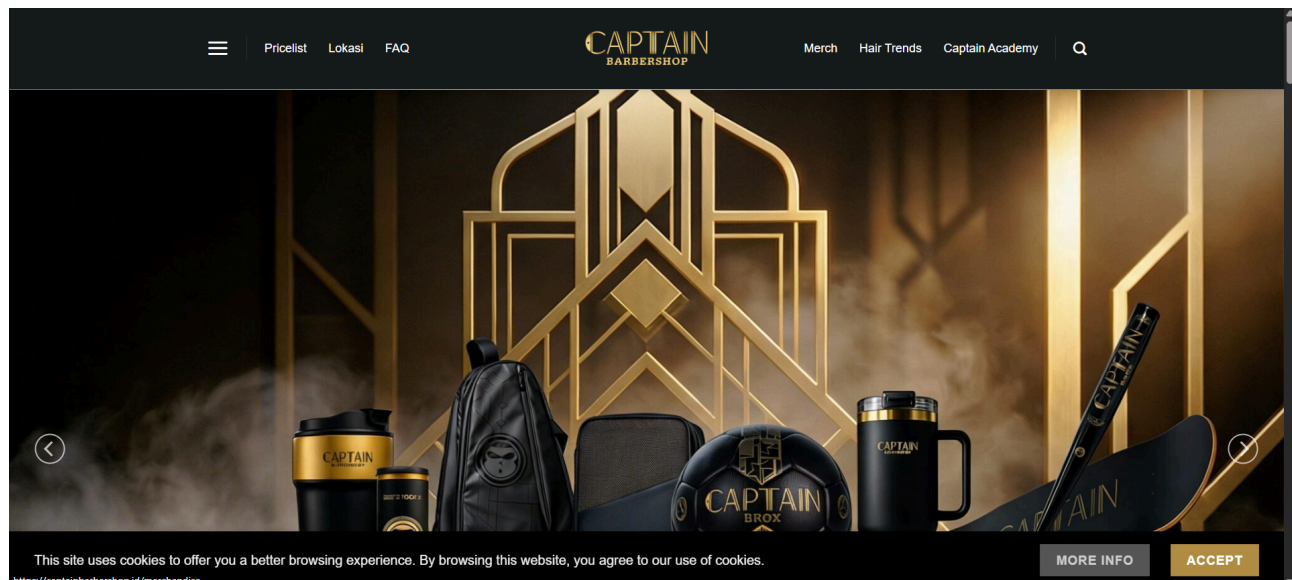
Customer Features:

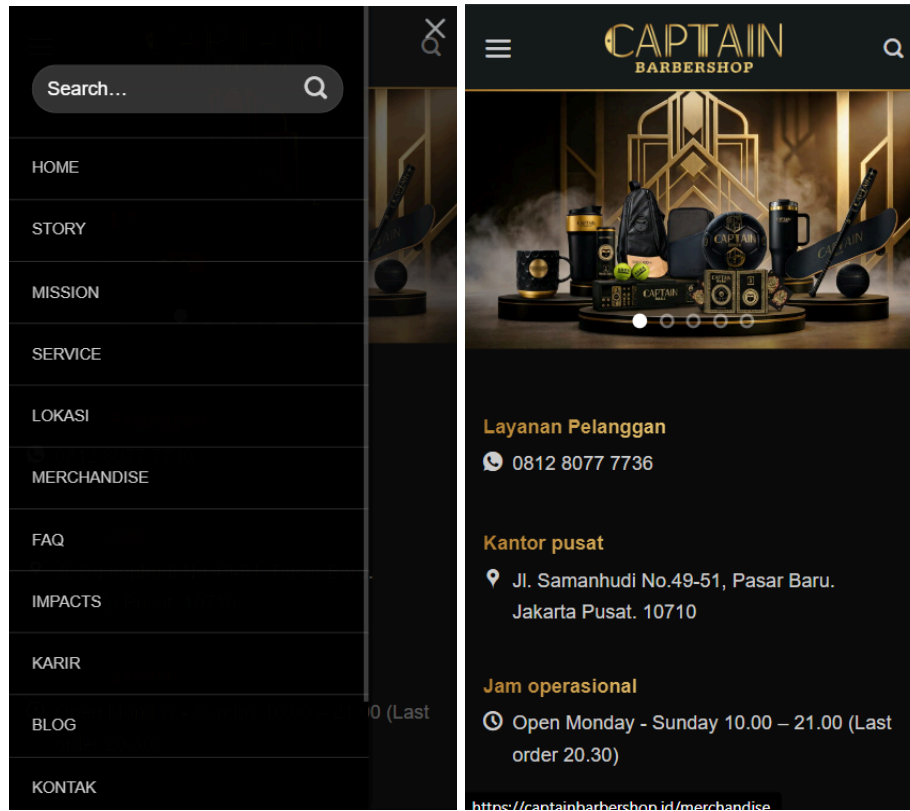
1. Dashboard with a view of available barber schedules.
2. Detailed barber information (rating, price, and years of experience).
3. Booking system based on available time slots.
4. "3-life" system with automatic deduction if late > 20 minutes.
5. Booking history.
6. User account settings.

Admin Features:

1. Customer account management.
2. Barber data management (profile, price, experience).
3. Barber schedule configuration.
4. Monitoring of active bookings and tardiness history.
5. Blocking and unblocking customer accounts.

1.5 References





Link towards reference : <https://captainbarbershop.id/>

2. Overall Description

2.1 Product Perspective

BarberHub is a new, self-contained web application prototype designed to demonstrate and validate the user workflow of a modern online barbershop reservation platform. The system is not a component of a larger enterprise system; it operates independently with its own database, authentication module, and scheduling engine.

The platform originates from a university project aimed at solving real-world inefficiencies in walk-in barbershop operations, specifically the problem of long unpredictable wait times, no-show customers, and the absence of transparent barber quality indicators. BarberHub provides a centralized, digital solution to these pain points by enabling online appointment booking, real-time schedule visibility, and automated discipline enforcement through its unique life system. The system interacts with two main user classes Customers and Admins through a single browser-based interface, backed by a server-side application and a relational database.

2.2 Product Functions

The major functions of the BarberHub system include:

- **User Authentication & Registration:** Allowing Customers and Admins to securely register and log in to the platform.
- **Barber Dashboard:** Displaying all active barbers with their real-time availability, rating, price, and years of experience, enabling Customers to browse and select a preferred barber.
- **Appointment Booking System:** Enabling Customers to select a barber and book an available time slot, with the system recording booking details and preventing double-booking of the same slot.
- **Life & Punctuality Enforcement System:** Automatically tracking Customer tardiness. If a Customer arrives more than 20 minutes late, one life is deducted. Account blocking is triggered automatically when all three lives are exhausted.
- **Rating & Review System:** Allowing Customers to submit a rating and an optional review for a barber after a completed appointment, which then updates the barber's overall displayed rating.
- **Booking History:** Providing both Customers and Admins with a chronological record of all past bookings, including statuses such as Completed, Late, or Cancelled.
- **Admin Management Panel:** A dedicated interface for Admins to manage barber profiles, configure schedules, monitor bookings, and perform account moderation (blocking/unblocking).

2.3 User Classes and Characteristics

The system identifies two primary user classes:

Customer

The primary and most frequent users of the application. Customers are individuals who wish to book haircut appointments online. They interact with the barber discovery dashboard, the booking system, and their booking history. Customers begin with 3 lives upon account registration and are subject to the punctuality enforcement system. They are expected to have basic digital literacy — the ability to navigate a web application on a smartphone or desktop browser.

Admin

Users with elevated system privileges. Admins are typically barbershop owners or managers. They are responsible for managing barber profiles, configuring time slot availability, monitoring all active and historical bookings, and moderating customer accounts (including blocking and manual life restoration). Admins are not subject to the life system and cannot make customer-facing bookings.

2.4 Operating Environment

BarberHub will operate as a responsive web application, accessible via modern web browsers on both desktop and mobile devices. The system is designed to be used on the go, particularly by Customers booking appointments from their smartphones.

Component	Specification
-----------	---------------

Client Interface	Responsive web application (desktop & mobile browsers)
Supported Browsers	Latest 2 versions of Google Chrome, Mozilla Firefox, Safari (iOS), Samsung Internet
Backend Runtime	Node.js (e.g., Express.js framework)
Database	Relational database (MySQL 8.0+ or PostgreSQL 13+)
Hosting (Prototype)	Local machine (XAMPP / Docker) or free-tier cloud (e.g., Railway, Supabase)

2.5 Design and Implementation Constraints

- **Web-Only Scope:** The prototype is limited to a browser-based interface. Native iOS or Android application development is currently out of scope.
- **Single Location:** The system is designed to serve a single barbershop. Multi-branch or franchise management is out of scope.
- **No Payment Gateway:** Online payment processing is not included in this prototype. Payments are assumed to be handled in-person at the barbershop.
- **Life System Logic:** Tardiness detection relies on Admin manual check-in confirmation, not GPS or automated arrival tracking. The system marks a booking as 'Late' when the Admin updates arrival status after the 20-minute grace window has passed.
- **Documentation Conventions:** All high-level requirements must be identified with a unique ID. UI elements must be bolded and User Roles italicized in all technical documentation, as described in Section 1.2.

2.6 User Documentation

The BarberHub project will deliver the following documentation components:

- **Software Requirements Specification (SRS):** This document, detailing all functional and non-functional requirements.
- **User Interface Specification:** A separate document or Figma prototype containing detailed UI designs and screen layouts for all major flows.
- **Admin Operations Guide:** A guide explaining how to manage barber profiles, configure schedules, update booking statuses, and moderate customer accounts.

2.7 Assumptions and Dependencies

The following assumptions and external dependencies apply to the BarberHub system:

Assumptions:

- All Customers are assumed to have a valid email address for registration and notification purposes.
- Customers are assumed to have access to a device with an internet connection and a modern web browser.

- The Admin is responsible for configuring all barber schedules before the system goes live. The system does not self-populate availability.
- A barber's time slot duration is fixed at 60 minutes per booking, regardless of service type.
- A Customer can only hold one active (upcoming) booking at a time. Concurrent bookings are not permitted.

Dependencies:

- The system depends on a reliable internet connection for real-time schedule updates and booking confirmations.
- Email notification functionality depends on a third-party email service provider
- Barber profile ratings depend on Customers completing post-appointment reviews. The rating cannot be updated if no review is submitted.

3. External Interface Requirements

3.1 User Interfaces

The BarberHub interface is designed to be clean, intuitive, and consistent across both desktop and mobile viewports. Because Customers may use the application on smartphones while on the go, the UI must prioritize clarity, thumb-reach on mobile, and fast navigation between booking flows.

GUI Standards and Product Style Guide:

- **Design Language:** The interface shall follow a modern, card-based design language inspired by Material Design 3 principles, adapted for a barbershop aesthetic.
- **Color Palette:** Primary actions (Book Now, Confirm) shall use the primary brand color (Dark Blue / Charcoal). Destructive or negative actions (Cancel Booking) shall use standard alert colors (Red). The life indicator shall use a distinctive color coding: green (3 lives), amber (2 lives), red (1 life).
- **Typography:** Sans-serif fonts (Inter or Poppins) must be used globally for legibility at various screen sizes.

Screen Layout Constraints:

- **Responsive Design:** All interfaces must be responsive across a minimum viewport width of 320px (mobile) up to 1440px (desktop), using CSS flexbox or grid layouts.
- **Navigation:** A persistent top navigation bar for desktop and a Bottom Navigation Bar for mobile shall provide access to Home/Dashboard, My Bookings, and My Account tabs for Customers.

Standard Buttons and Functions:

- **Global Booking CTA:** A prominent Book Now button shall appear on each barber card on the dashboard.
- **Life Status Indicator:** A persistent life indicator (3 heart/shield icons) shall appear in the Customer's account header at all times.

Error Message Display Standards:

- All system errors, validation failures, and success states shall be displayed as Toast Notifications at the bottom of the screen, auto-dismissing after 3–4 seconds.
- **Success:** Green background with a checkmark icon ("Booking Confirmed!").
- **Error/Warning:** Red background with an alert icon ("This time slot is no longer available.").

Software Components Requiring a UI:

- **Authentication Module:** Login screen, Registration screen (with email, password, and full name), and Forgot Password screen.
- **Customer Dashboard:** Barber listing with cards showing profile picture, name, rating, price, and experience. Includes a filter/sort bar.
- **Barber Profile & Booking View:** Detailed barber page with available time slot calendar. The customer selects a slot and taps Book Now.
- **My Bookings View:** A list of upcoming and past bookings with status badges (Upcoming, Completed, Late, Cancelled).
- **Post-Appointment Review Modal:** A star rating widget and optional text field, triggered after a booking is marked as Completed.
- **Admin Management Console:** A table-based interface for managing barbers, schedules, booking oversight, and customer account moderation.

3.2 Hardware Interfaces

BarberHub is a web application and does not require proprietary external hardware. It operates on standard consumer-grade devices (smartphones, tablets, laptops, and desktops) through their web browsers.

- **Supported Form Factors:** Smartphones (primary), tablets, laptops, desktop computers.
- **Minimum Device Requirements:** Any device capable of running a modern web browser with an active internet connection. No specific hardware peripherals (such as cameras) are required for core booking functionality.
- **Network Hardware:** The system depends on the host device's network interface (Wi-Fi or Cellular data, minimum 3G) to communicate with the backend server. All traffic is routed over HTTPS.

3.3 Software Interfaces

The BarberHub prototype will utilize a modern web stack:

Layer	Technology/Interface
Frontend Framework	React.js or Vue.js handles UI state, routing, and barber schedule rendering
Backend / API Layer	Node.js with Express.js handles business logic, authentication, and database queries
Database Management System	MySQL 8.0+ or PostgreSQL 13+ stores users, barbers, schedules, bookings, and reviews
Authentication	JSON Web Tokens (JWT) issued on login and validated on every protected API request
Email Notification Service	Third-party SMTP provider (SendGrid, Nodemailer) sends

	booking confirmation and reminder emails
Data Exchange Format	JSON (JavaScript Object Notation) used for all client-server API communication

3.4 Communications Interfaces

- **Protocol:** All client-server communication shall occur over HTTPS (HTTP Secure, TLS 1.2 or higher) to protect user data in transit.
- **Architecture:** The system shall use a RESTful API architecture. Standard HTTP methods must be applied semantically: GET for fetching barber and schedule data, POST for creating bookings and user accounts, PUT/PATCH for updating booking status and barber profiles, and DELETE for removing cancelled bookings or barber records.
- **Real-Time Availability:** Time slot availability shall be refreshed via standard REST polling on the booking page. The slot display must reflect confirmed bookings within a maximum latency of 5 seconds.

4. System Features

This section outlines the core functional requirements of the BarberHub prototype, categorized by the major services provided by the platform.

4.1 User Authentication & Registration

4.1.1 Description and Priority

Priority: High. This feature forms the entry point of the system. All core functionality booking, the life system, and booking history — is tied to an authenticated Customer account. Registration is required before any booking can be made.

4.1.2 Stimulus/Response Sequences

- **Stimulus:** A new user navigates to the BarberHub application and taps Register.
- **System Response:** The system presents a registration form requesting Full Name, Email Address, and Password.
- **Stimulus:** User submits the form with valid data.
- **System Response:** The system creates the account, initializes the customer's life count to 3, and redirects to the Customer Dashboard. A confirmation email is sent to the registered address.
- **Stimulus:** An existing user enters their credentials on the Login screen.
- **System Response:** The system validates the credentials, issues a JWT session token, and routes the user to the Dashboard.
- **Stimulus:** A user submits invalid credentials.
- **System Response:** The system displays a red Toast notification stating "Invalid email or password" and clears the password field.
- **Stimulus:** A blocked Customer attempts to log in.
- **System Response:** The system displays a message stating "Your account has been suspended. Please contact the barbershop." and prevent access

4.1.3 Functional Requirements

- REQ-1.1:** The system shall require an email address and password to register a new Customer account.
- REQ-1.2:** The system shall initialize every newly registered Customer account with a life count of exactly 3.
- REQ-1.3:** Passwords must be stored as a hashed value using the crypt algorithm with a minimum of 10 salt rounds. Plaintext passwords must never be stored or logged.
- REQ-1.4:** The system shall authenticate users via JWT, with a token expiration time of 24 hours.
- REQ-1.5:** The system shall enforce Role-Based Access Control (RBAC): Customer accounts have access to booking and review flows; Admin accounts have access to the management console.
- REQ-1.6:** The system shall prevent login for accounts with a blocked status and display an informative error message.

REQ-1.7: The system shall provide a "Forgot Password" feature that generates a secure, time-limited password reset token (expiring in 15 minutes) sent to the Customer's registered email address.

4.2 Barber Dashboard

4.2.1 Description and Priority

Priority: High. The dashboard is the primary screen for authenticated Customers. It displays all active barbers with key information to help customers choose who to book with.

4.2.2 Stimulus/Response Sequences

- **Stimulus:** An authenticated Customer logs in and is routed to the Dashboard.
- **System Response:** The system fetches and displays a list of all active barber profiles as cards, each showing the barber's name, profile photo, rating, service price, and years of experience.
- **Stimulus:** Customer uses the Sort By filter (for example: "Highest Rating", "Lowest Price", "Most Experienced").
- **System Response:** The dashboard re-renders the barber cards in the selected order without a full page reload.
- **Stimulus:** Customer taps a barber card.
- **System Response:** The system navigates to the Barber Profile & Booking View for that specific barber.

4.2.3 Functional Requirements

REQ-2.1: The system shall display all barbers with status 'Active' on the Customer Dashboard.

REQ-2.2: Each barber card shall display: profile photo (or a default placeholder), full name, overall rating (displayed as a star score out of 5), service price, and years of experience.

REQ-2.3: The system shall provide sorting options for the barber list: by Rating (descending), by Price (ascending), and by Experience (descending).

REQ-2.4: The system shall display each barber's real-time availability status (e.g., 'Available Today' or 'Next available: [date]') on the barber card.

4.3 Appointment Booking System

4.3.1 Description and Priority

Priority: High. This is the core transactional feature of BarberHub. It allows Customers to reserve a specific time slot with their chosen barber.

4.3.2 Stimulus/Response Sequences

- **Stimulus:** Customer navigates to a Barber Profile page.

- **System Response:** The system displays the barber's available time slots in a weekly calendar view, distinguishing between Available (bookable), Booked (taken), and Unavailable (outside working hours) slots.
- **Stimulus:** Customer selects an available time slot and taps Book Now.
- **System Response:** The system validates that the slot is still available and that the Customer does not have an existing active booking. If valid, a confirmation modal appears summarizing the booking details (barber name, date, time, price).
- **Stimulus:** Customer taps Confirm Booking in the confirmation modal.
- **System Response:** The system creates the booking record, marks the slot as Booked, sends a confirmation email to the Customer, and displays a green Toast notification "Booking Confirmed!".
- **Stimulus:** Customer attempts to book a slot that was just taken by another user.
- **System Response:** The system rejects the booking and displays an error Toast: "This time slot is no longer available. Please select another slot."
- **Stimulus:** Customer attempts to cancel an upcoming booking from My Bookings view.
- **System Response:** A cancellation confirmation dialog appears. Upon confirmation, the booking is cancelled, the slot is reopened, and the Customer is notified via Toast.

4.3.3 Functional Requirements

- REQ-3.1:** The system shall prevent a Customer from booking a time slot that is already marked as Booked.
- REQ-3.2:** The system shall prevent a Customer from holding more than one active (Upcoming) booking at any time.
- REQ-3.3:** Upon successful booking, the system shall send a confirmation email to the Customer containing the barber name, date, time, and estimated price.
- REQ-3.4:** The system shall allow a Customer to cancel an Upcoming booking. Cancellation releases the time slot back to available status.
- REQ-3.5:** The system shall record a timestamp for when the booking was created and when the Admin confirms the Customer's arrival.
- REQ-3.6:** The system must implement database-level locking during the booking transaction to prevent race conditions. If multiple users attempt to book the same time slot simultaneously, the system shall only fulfill the first request and return an error for subsequent requests.

4.4 Life & Punctuality Enforcement System

4.4.1 Description and Priority

Priority: High. This is the distinguishing feature of BarberHub. It creates accountability among Customers to honor their bookings and arrive on time, protecting barbers from idle time caused by late or absent customers.

4.4.2 Stimulus/Response Sequences

- **Stimulus:** A Customer's appointment time passes. The Admin checks whether the Customer has arrived.

- System Response (On-Time / Early): Admin marks the booking as Customer Arrived within the 20-minute grace window. No life is deducted.
- System Response (Late — > 20 min): Admin marks the Customer as Arrived Late, or marks them as a No-Show after the window closes. The system automatically deducts 1 life from the Customer's account.
- System Response (Life Count = 2 or 1): The system sends a warning email to the Customer informing them of their remaining lives.
- System Response (Life Count = 0): The system immediately blocks the Customer's account, preventing any future logins or bookings. A notification email is sent to the Customer.
- **Stimulus:** Admin navigates to the Customer management panel and manually restores a blocked Customer's account or resets their life count.
- System Response: The system updates the Customer's life count and account status as specified by the Admin. The change is logged in the audit trail.

4.4.3 Functional Requirements

- REQ-4.1:** Every new Customer account must be initialized with exactly 3 lives upon successful registration.
- REQ-4.2:** The system shall deduct exactly 1 life from a Customer's account when the Admin marks their booking status as 'Late' or 'No-Show'.
- REQ-4.3:** When a Customer's life count reaches 0, the system shall immediately and automatically set the account status to Blocked.
- REQ-4.4:** A Blocked Customer account shall be prevented from logging in, browsing, or booking any appointments.
- REQ-4.5:** The system shall send an automated email notification to the Customer whenever a life is deducted, and a separate notification when their account is blocked.
- REQ-4.6:** Only an Admin shall be able to manually restore a Customer's life count or unblock a Customer account.
- REQ-4.7:** All life deductions and account status changes must be logged with a timestamp and the performing user (system or admin) for audit purposes.
- REQ-4.8:** The system shall execute an automated daily reconciliation script (e.g., Cron Job) at 23:59 to identify any bookings in the past that are still marked as 'Upcoming'. These records must be flagged as 'Requires Admin Review' to prevent unfair life deductions or missed penalties.

4.5 Rating & Review System

4.5.1 Description and Priority

Priority: Medium. This feature allows Customers to provide feedback on their barber experience, which in turn improves the quality and usefulness of the barber sorting and discovery features on the dashboard.

4.5.2 Stimulus/Response Sequences

- **Stimulus:** A booking is marked as Completed by the Admin.

- **System Response:** The system triggers a post-appointment review prompt for the Customer (displayed in the My Bookings view or via email).
- **Stimulus:** Customer opens the review prompt and submits a star rating (1–5) and an optional text comment.
- **System Response:** The system stores the review, recalculates the barber's overall rating (average of all ratings received), and updates the displayed rating on the barber's card and profile page.

4.5.3 Functional Requirements

- REQ-5.1:** The system shall allow a Customer to submit one rating per completed booking. Re-rating the same booking is not permitted.
- REQ-5.2:** Ratings shall be integers on a scale of 1 to 5 (1 = Poor, 5 = Excellent). The text review field is optional.
- REQ-5.3:** The system shall recalculate and update the barber's overall rating (displayed rating = average of all submitted ratings, rounded to one decimal place) immediately upon review submission.
- REQ-5.4:** The review option shall only be available for bookings with a status of Completed. Customers cannot review bookings marked as Late, No-Show, or Cancelled.

4.6 Booking History & Status Tracking

4.6.1 Description and Priority

Priority: Medium. Provides Customers and Admins with a transparent, chronological record of all booking activity, supporting accountability and dispute resolution.

4.6.2 Stimulus/Response Sequences

- **Stimulus:** Customer navigates to the My Bookings tab.
- **System Response:** The system displays a list of all the Customer's bookings (past and upcoming), sorted by date (newest first), each showing: barber name, date and time slot, booking status badge, and (if applicable) the Customer's submitted rating.
- **Stimulus:** Admin navigates to the Booking Monitor panel.
- **System Response:** The system displays all bookings across all Customers, filterable by date, barber, and status.

4.6.3 Functional Requirements

- REQ-6.1:** The system shall record and display the following booking statuses: Upcoming, Completed, Late, No-Show, and Cancelled.
- REQ-6.2:** A Customer shall only be able to view their own booking history. An Admin can view all bookings across all Customers.
- REQ-6.3:** The Admin booking monitor shall be filterable by Date Range, Barber, and Booking Status.

4.7 Admin Management Panel

4.7.1 Description and Priority

Priority: High. The Admin panel is the operational backbone of BarberHub. Without it, there is no way to configure barbers, manage schedules, or moderate customer accounts.

4.7.2 Stimulus/Response Sequences

- **Stimulus:** Admin navigates to Manage Barbers and clicks Add New Barber.
- **System Response:** A form is displayed requesting the barber's full name, profile photo, service price, and years of experience. The new barber is set to Active status by default.
- **Stimulus:** Admin navigates to Manage Schedules for a specific barber and configures working days and time slot availability.
- **System Response:** The system generates bookable time slots for the selected barber based on the configured working hours, in 60-minute intervals.
- **Stimulus:** Admin navigates to Customer Accounts, selects a Customer, and clicks Block Account or Restore Life.
- **System Response:** The system updates the Customer's account status or life count as directed. The action is logged with a timestamp.

4.7.3 Functional Requirements

- REQ-7.1:** The system shall allow the Admin to create, update, and deactivate barber profiles. Deactivating a barber hides them from the Customer Dashboard without deleting historical booking records.
- REQ-7.2:** The system shall allow the Admin to configure each barber's available working days and hours, from which the system auto-generates bookable 60-minute time slots.
- REQ-7.3:** The system shall allow the Admin to update any booking's status (to Completed, Late, or No-Show) from the booking monitor panel.
- REQ-7.4:** The system shall allow the Admin to block or unblock a Customer account, and to manually adjust a Customer's life count (within the 0–3 range).
- REQ-7.5:** All Admin actions on Customer accounts (blocking, unblocking, life restoration) must be persisted as audit log entries, recording the Admin's identity, the action taken, the target Customer, and a timestamp.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **API Response Latency:** All REST API responses for standard operations (fetching barber lists, creating bookings, updating statuses) must complete within 500 milliseconds under normal load conditions.
- **Dashboard Load Time:** The initial Customer Dashboard, including barber cards and availability indicators, must render within 2 seconds on a standard 4G mobile network.
- **Booking Slot Freshness:** Time slot availability displayed on the booking page must reflect the actual database state within a maximum of 5 seconds.
- **Application Payload:** The initial web application payload (HTML/CSS/JS) must be under 3MB to ensure acceptable load times across mobile networks.
- **Concurrent Users:** The system must support a minimum of 50 concurrent users without degraded performance for the prototype scope.
- **List Pagination:** To ensure the Application Payload remains lightweight, all endpoints returning lists of data (e.g., Booking History, Barber Lists, Admin Booking Monitor) must implement pagination or infinite scrolling, fetching a maximum of 20 records per request.

5.2 Safety Requirements

- **Data Loss Prevention:** The booking system must validate slot availability in the database immediately before committing a booking transaction, using database-level locking or transactions to prevent race conditions where two Customers simultaneously book the same slot.
- **Life Deduction Idempotency:** The system must ensure that a life deduction event for a single booking is only applied once, even if the Admin action is triggered or submitted multiple times.

5.3 Security Requirements

- **Encryption in Transit:** All client-server communication must be encrypted using TLS 1.2 or higher (HTTPS). HTTP requests must be forcefully redirected to HTTPS.
- **Authentication Mechanism:** User sessions must be managed using JSON Web Tokens (JWT). Tokens must have a maximum expiration of 24 hours to limit exposure from unattended sessions.
- **Password Security:** User passwords must be hashed using the bcrypt algorithm with a minimum of 10 salt rounds. Plaintext passwords must never be stored, logged, or transmitted.
- **Injection Prevention:** All database queries must use parameterized statements or an ORM to mitigate SQL injection vulnerabilities.
- **Role Enforcement:** RBAC must be enforced both on the client-side (hiding Admin UI elements from Customers) and on the backend API (rejecting unauthorized API calls with HTTP 403 Forbidden responses).
- **Input Validation:** All user inputs (registration fields, review text) must be validated and sanitized on the server side to prevent XSS and injection attacks.

5.4 Software Quality Attributes

- **Usability:** The Customer booking flow (from Dashboard to Confirmed Booking) must be completable in no more than 4 navigation steps/taps. All primary action buttons must have a minimum touch target size of 44x44 CSS pixels.
- **Reliability:** If the client device loses network connectivity during a booking attempt, the system must detect the failure, abort the booking transaction, and display a 'Network Error, Booking not confirmed' message. The system must not create partial booking records.
- **Maintainability:** The codebase must follow a modular architecture (separation of frontend, backend, and database layers) to allow individual components to be updated without requiring full system redeployment.
- **Scalability:** The database schema must be designed to accommodate future features such as multiple barbershop locations and service catalog expansion without requiring a full schema redesign.
- **Accessibility:** The UI must maintain a minimum contrast ratio of 4.5:1 for all body text against its background (WCAG AA standard) to ensure readability across devices and lighting conditions.

5.5 Business Rules

- **Rule 1 - Life Floor:** A Customer's life count cannot fall below 0. The moment it reaches 0, account blocking is triggered. The system must not process further deductions on an already-blocked account.
- **Rule 2 - Life Ceiling:** A Customer's life count cannot exceed 3 at any time. Admin-initiated restorations must cap the value at 3.
- **Rule 3 - Booking Immutability:** Once a booking's status is set to Completed, Late, or No-Show by an Admin, that status cannot be changed. If an error was made, a new corrective log entry must be created.
- **Rule 4 - One Active Booking:** A Customer may hold no more than one booking with an Upcoming status at any time. The system must enforce this rule at the database level.
- **Rule 5 - Soft Deletion for Barbers:** Barber profiles cannot be hard-deleted if they are associated with any booking records. They must be set to Inactive status to preserve historical data integrity.
- **Rule 6 - No Retroactive Rating:** A Customer may only submit a review for a booking that is in Completed status. Reviews cannot be submitted, edited, or deleted after initial submission.

6. Other Requirements

6.1 Database Requirements

- **Relational Structure:** The system requires a relational database (MySQL 8.0+ or PostgreSQL 13+) to enforce referential integrity between core entities: Users, Barbers, Schedules, Bookings, Reviews, and Audit Logs.
- **Core Entities:** The database schema must include at minimum the following tables: users (with role, life_count, and is_blocked fields), barbers, time_slots, bookings (with status and timestamps), reviews, and admin_audit_log.
- **Local Prototyping:** For the academic prototype demonstration, the database may be hosted locally (via Docker or XAMPP) or on a lightweight free-tier cloud instance (Railway, Supabase, or a managed AWS RDS micro instance).
- **Data Integrity:** Foreign key constraints must be enforced across all relational tables. Cascading deletes are prohibited for booking and audit records; soft deletion (status flags) must be used instead.
- **Timezone Standardization:** All timestamps, scheduling data, and audit logs must be stored in the database strictly in UTC (Coordinated Universal Time). The client-side application is responsible for converting and displaying these times in the user's local timezone (e.g., WIB).

6.2 Browser & Device Support

The application must support the following environments

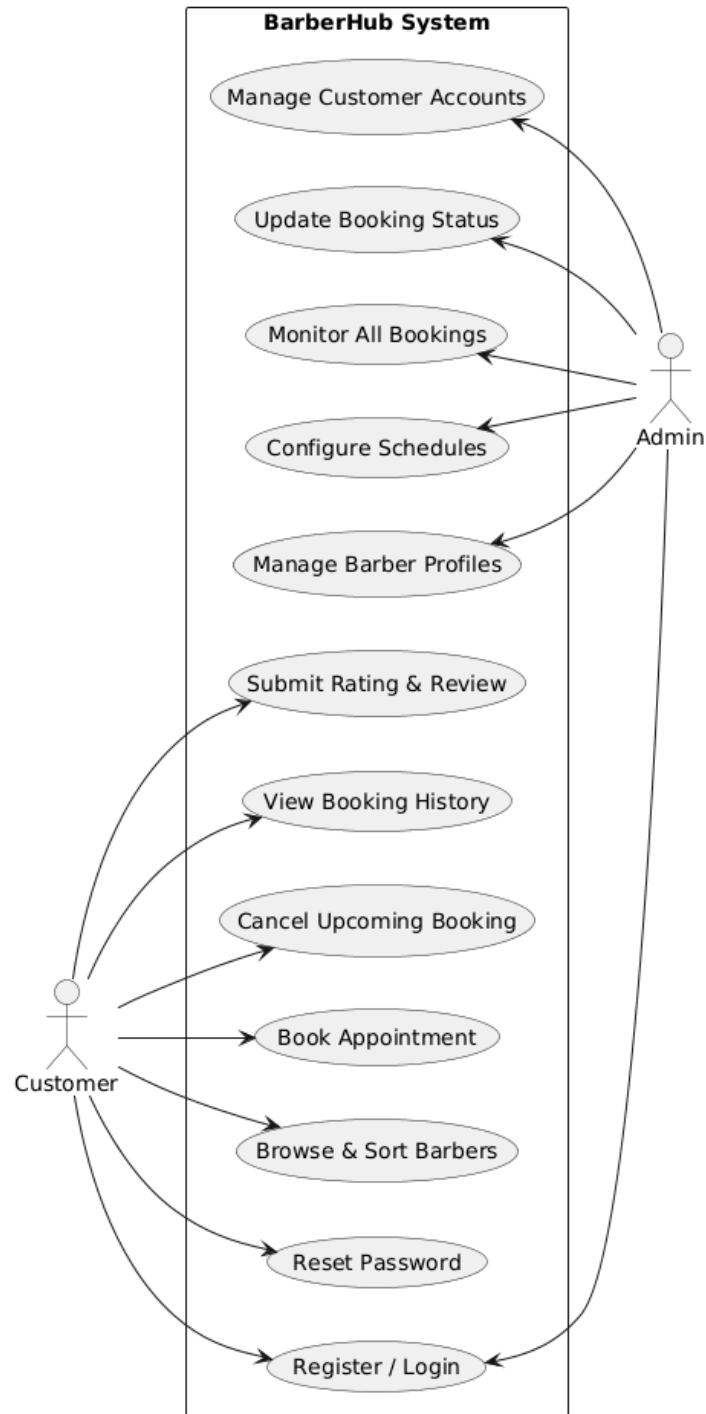
Platform	Minimum Supported Version
Google Chrome (Android & Desktop)	Latest 2 major versions
Mozilla Firefox (Desktop)	Latest 2 major versions
Apple Safari (iOS & macOS)	Latest 2 major versions
Samsung Internet (Android)	Latest 2 major versions
Microsoft Edge (Desktop)	Latest 2 major versions

Appendix A: Glossary

Term	Definition
JWT	JSON Web Token. A compact, URL-safe means of representing authentication claims between two parties, used here for session authorization.
RBAC	Role-Based Access Control. Restricting system access to authorized users based on their assigned role (Customer vs. Admin).
REST / RESTful API	Representational State Transfer. An architectural style for designing networked applications using standard HTTP methods.
CRUD	Create, Read, Update, Delete. The four fundamental operations of persistent data storage.
Life (Nyawa)	A discipline credit unit assigned to each Customer account (initialized at 3). Deducted when a Customer is late by more than 20 minutes. When all lives are depleted, the account is blocked.
Time Slot	A 60-minute bookable appointment window configured by the Admin for a specific barber on a specific date and time.
Soft Deletion	Marking a record as Inactive or Archived in the database rather than permanently removing it, preserving historical data integrity.
Booking Status	The current state of a booking record: Upcoming, Completed, Late, No-Show, or Cancelled.
HTTPS / TLS	HyperText Transfer Protocol Secure / Transport Layer Security. Protocols for encrypting data transmitted between client and server.
ORM	Object-Relational Mapper. A programming technique for converting data between incompatible type systems (object-oriented code and a relational database).
Toast Notification	A brief, non-intrusive UI notification that appears at the bottom or top of the screen and auto-dismisses after a few seconds.
WCAG AA	Web Content Accessibility Guidelines Level AA. A set of standards for web accessibility, including a 4.5:1 minimum contrast ratio for normal text.

7. Unified Modeling Languages

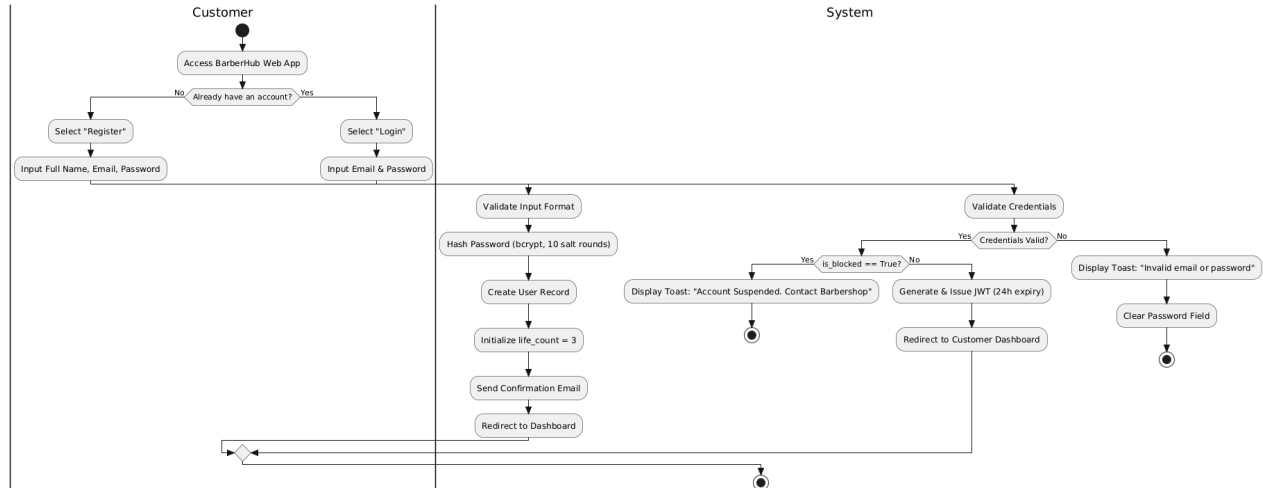
7.1 Use Case Diagram



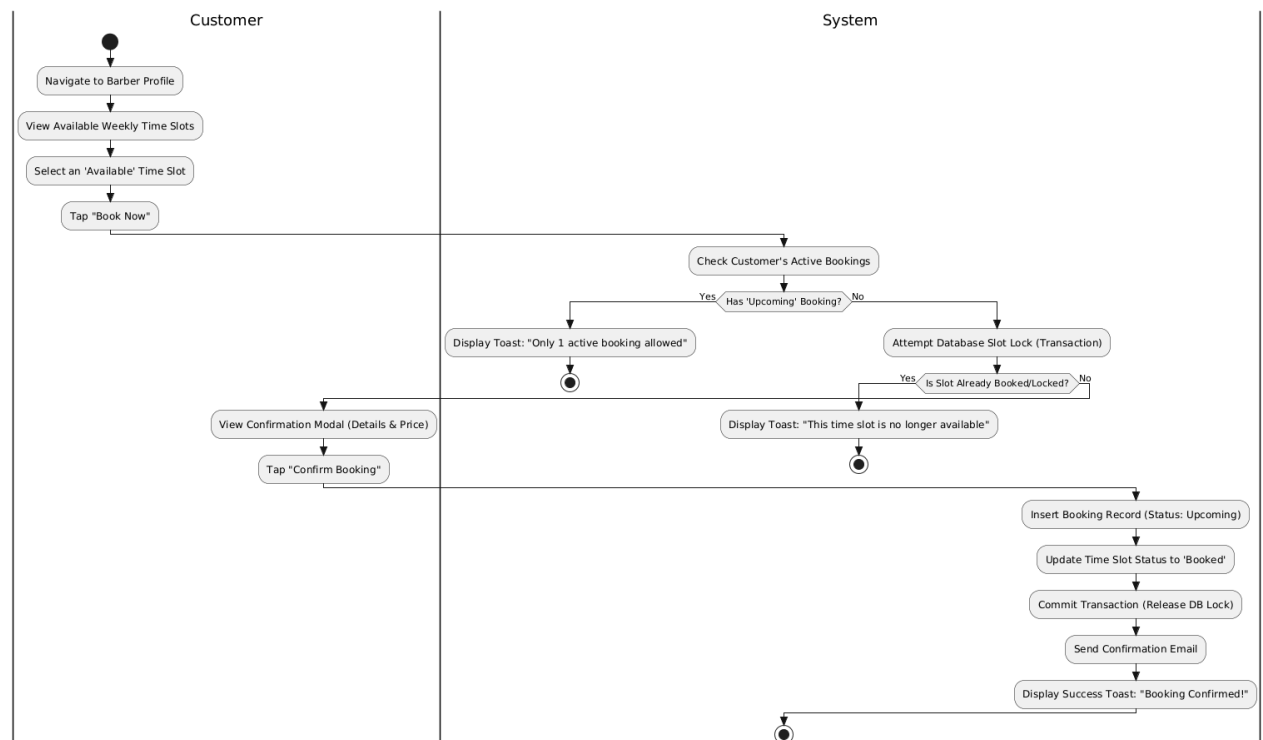
7.2 Activity Diagrams

For Users

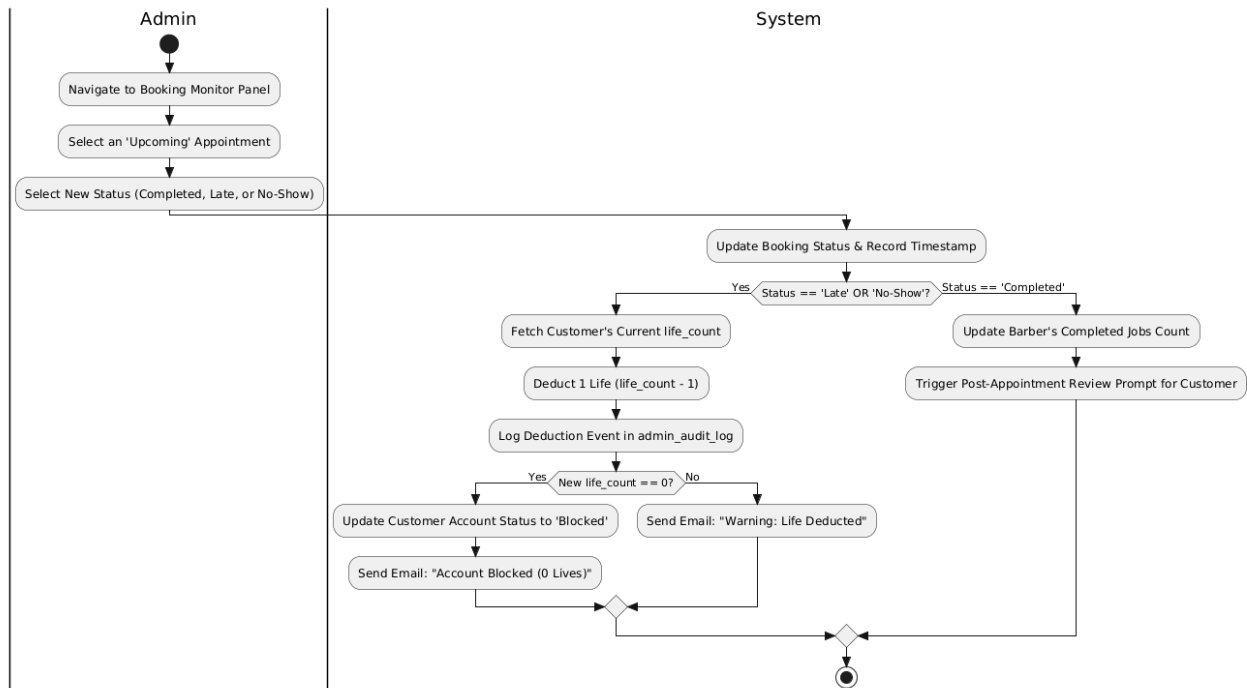
7.2.1 Use Case: Customer Registration & Login



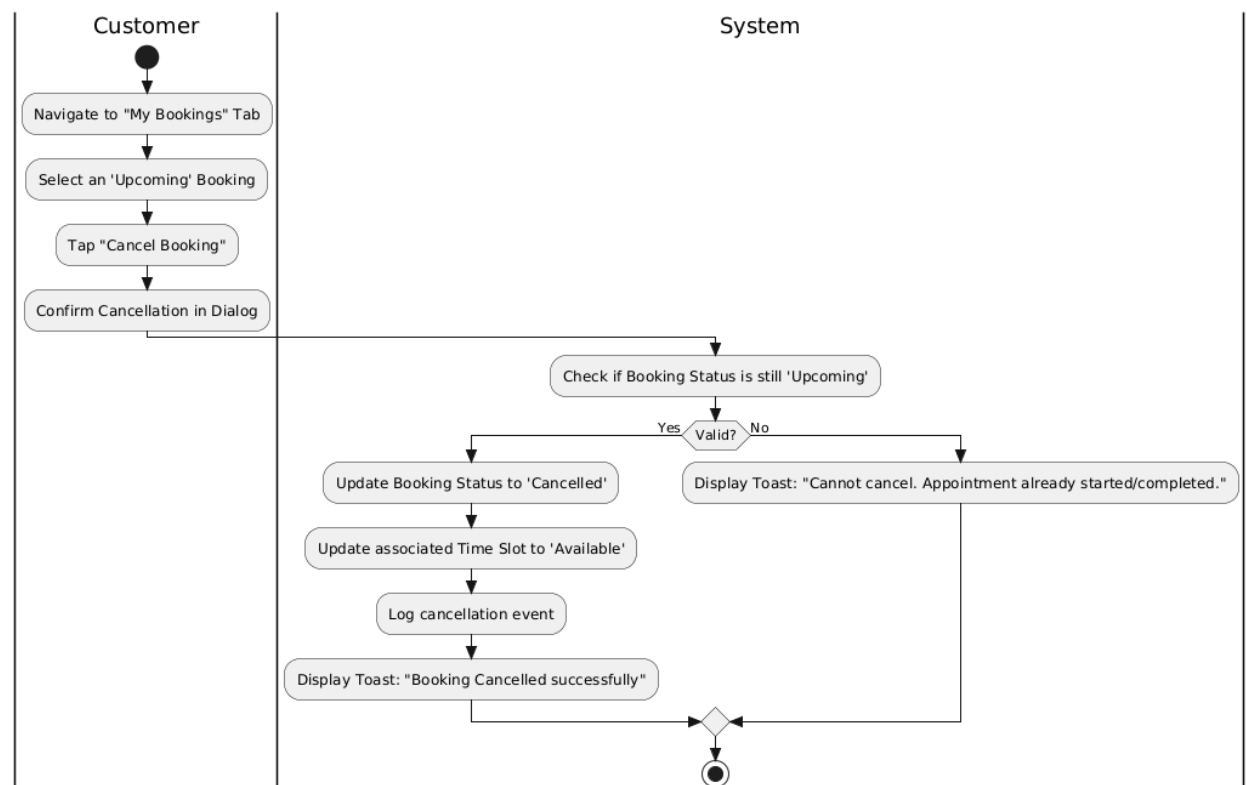
7.2.2 Use Case: Book Appointment (The Concurrency Flow)



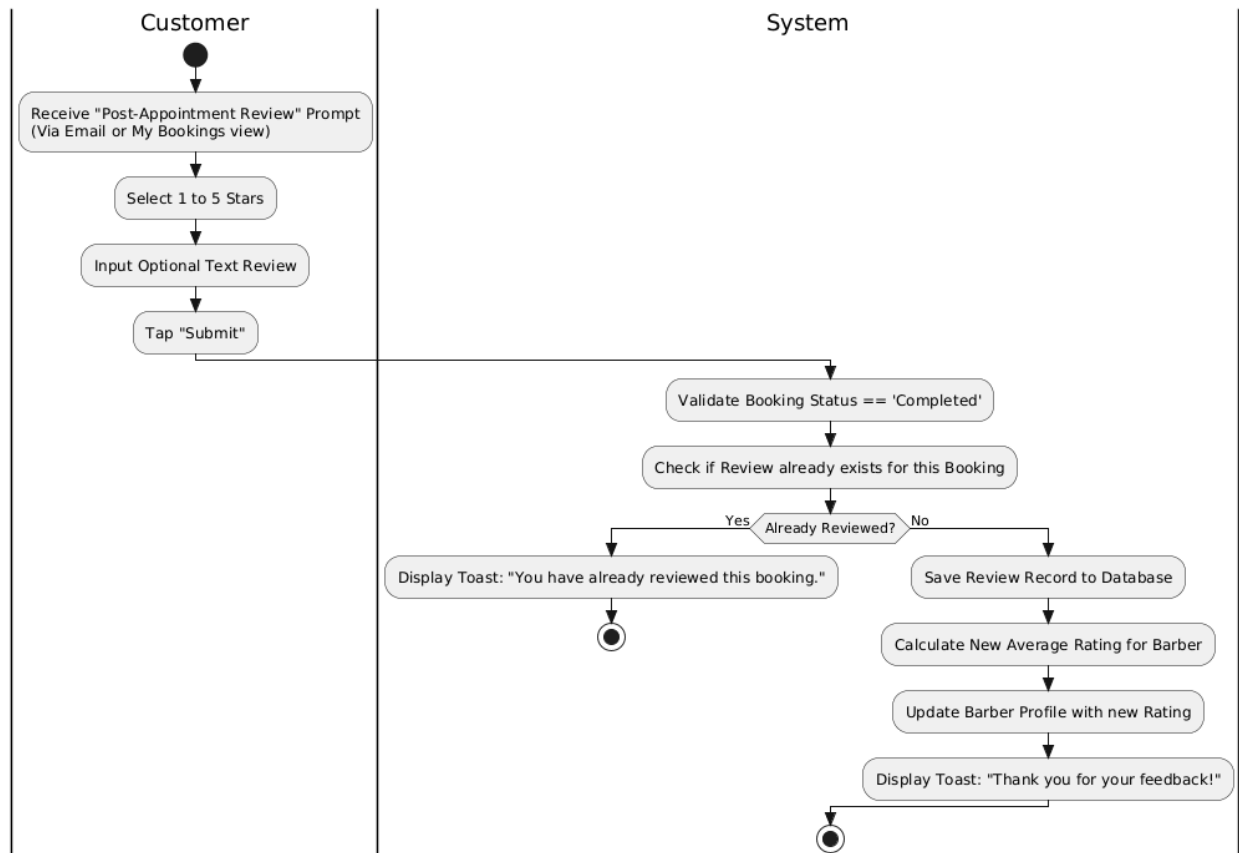
7.2.3 Use Case: Update Booking Status & Penalty Enforcement



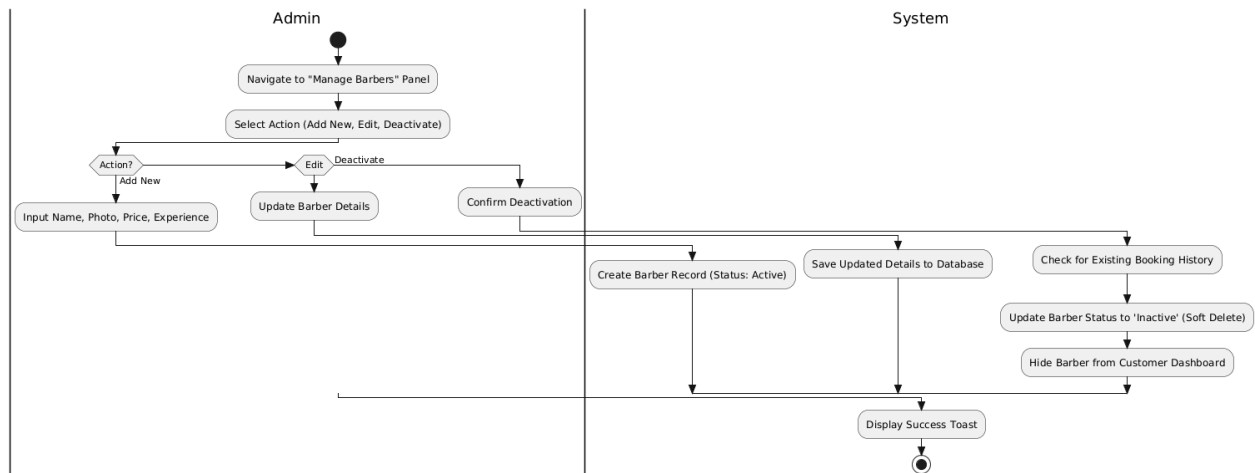
7.2.4 Use Case: Update Booking Status & Penalty Enforcement



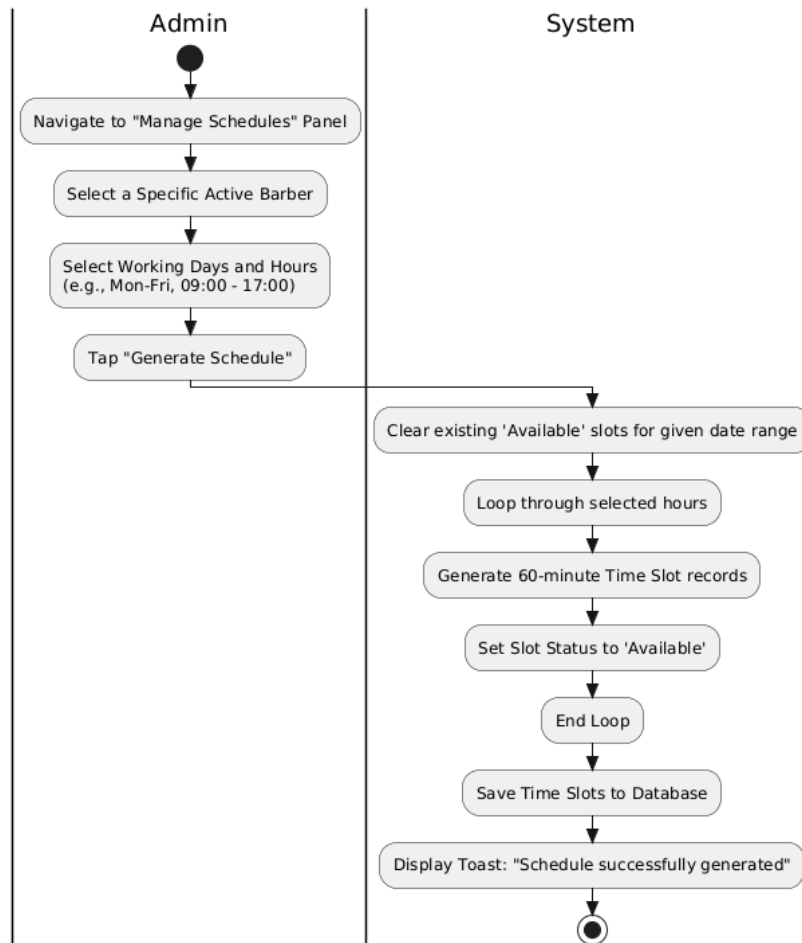
7.2.5 Use Case: Submit Rating & Review



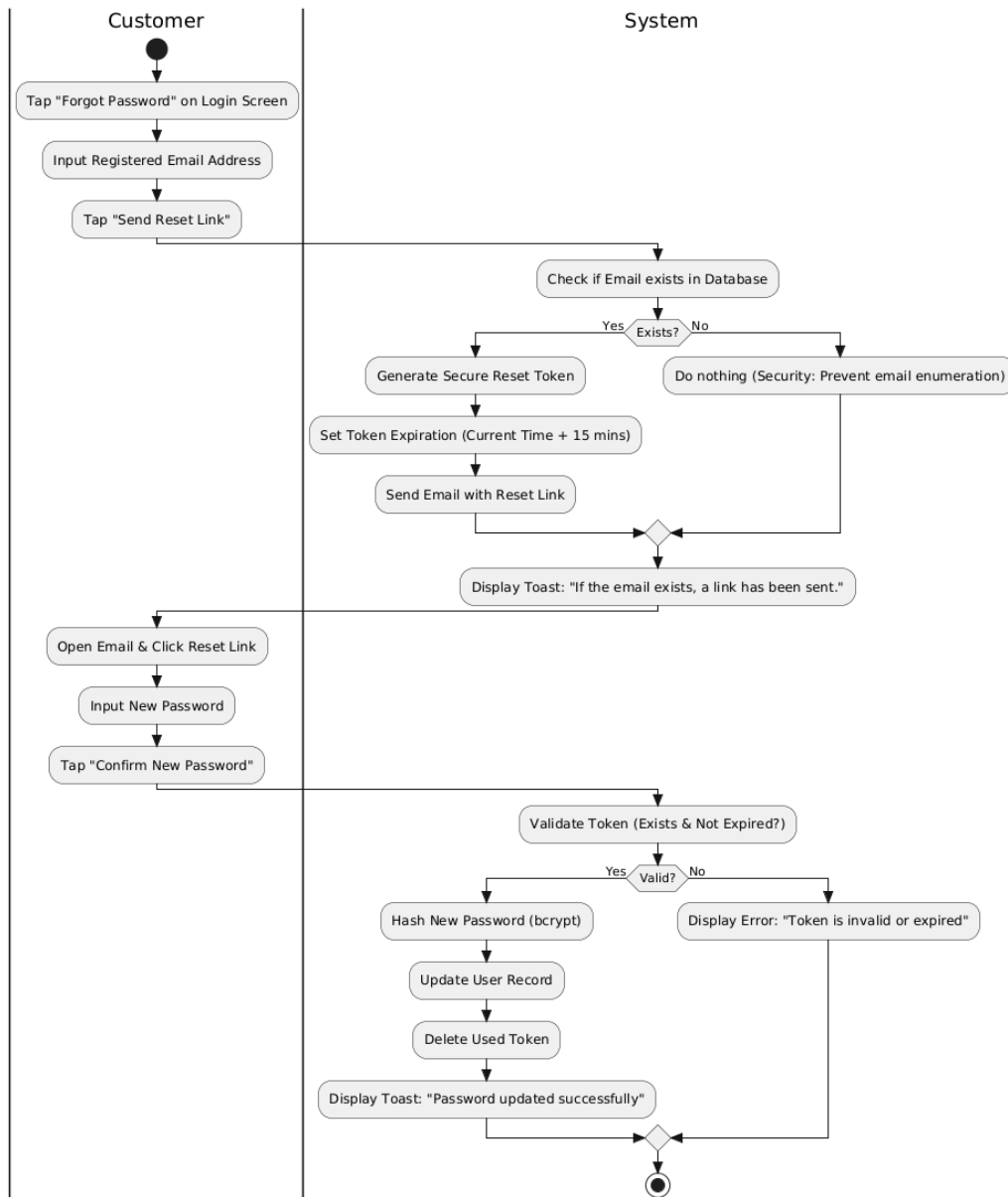
7.2.6 Use Case: Manage Barber Profiles



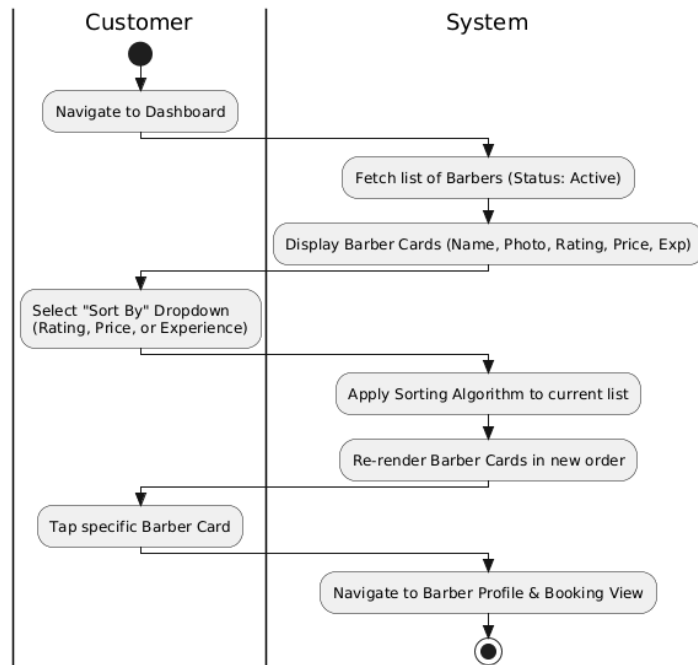
7.2.7 Use Case: Configure Schedules



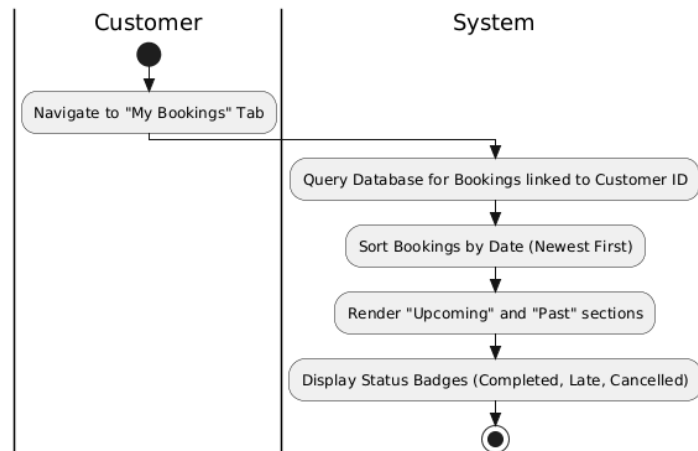
7.2.8 Use Case: Reset Password



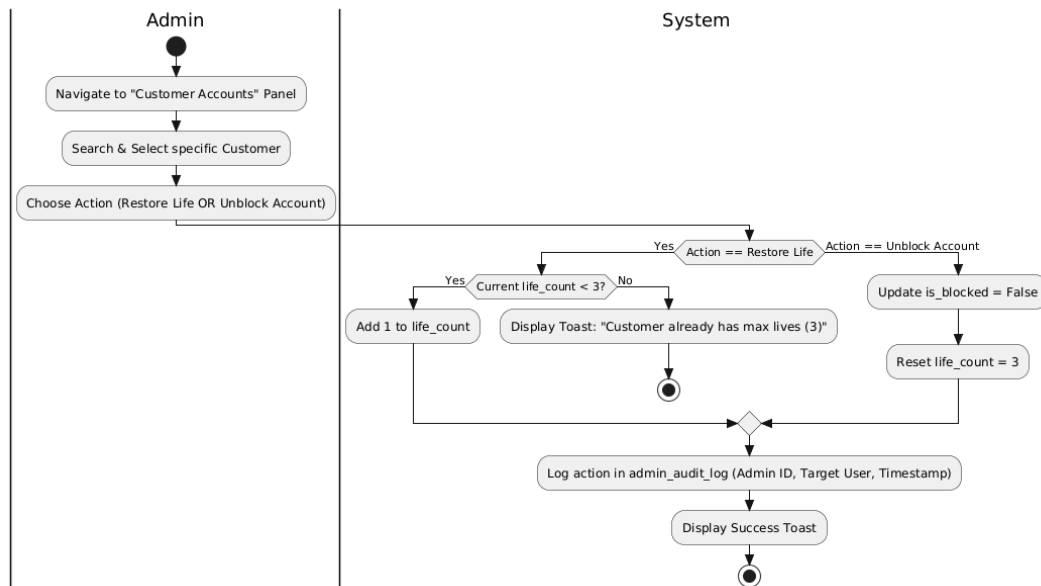
7.2.9 Use Case: Browse & Sort Barbers



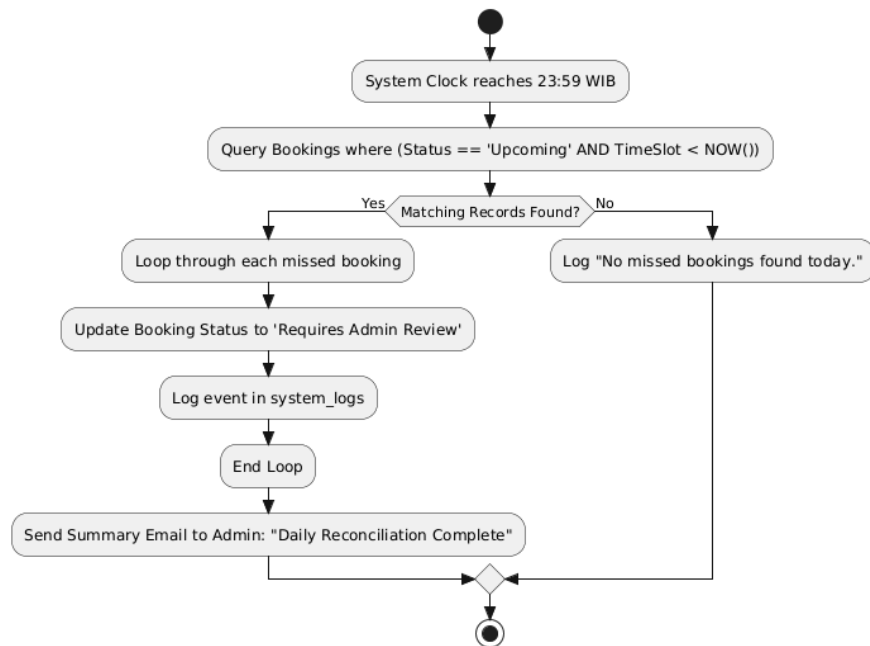
7.2.10 Use Case: View Booking History



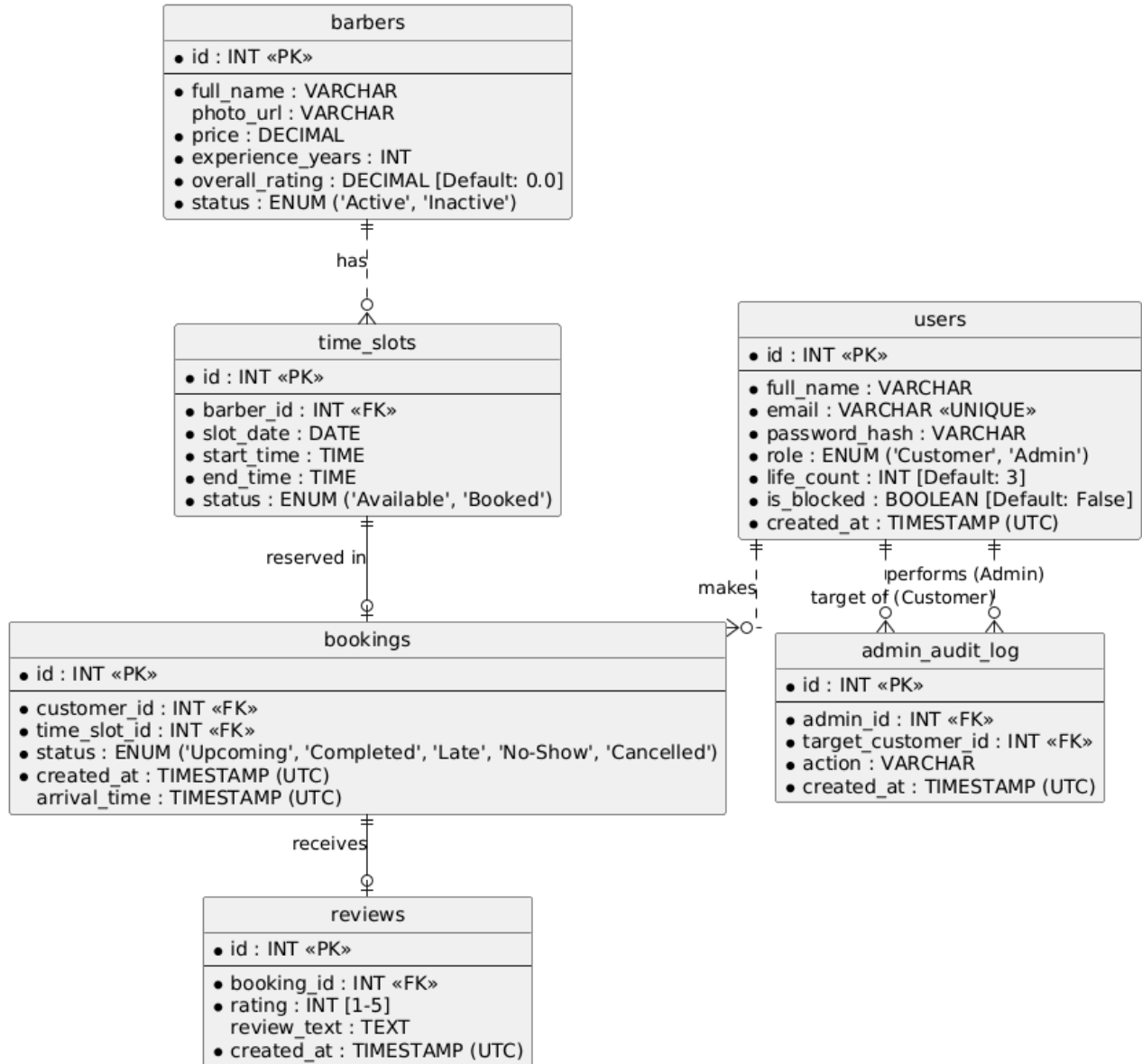
7.2.11 Use Case: View Booking History

**Automated by System**

7.2.12 The Automated Cron Job (End-of-Day Reconciliation)

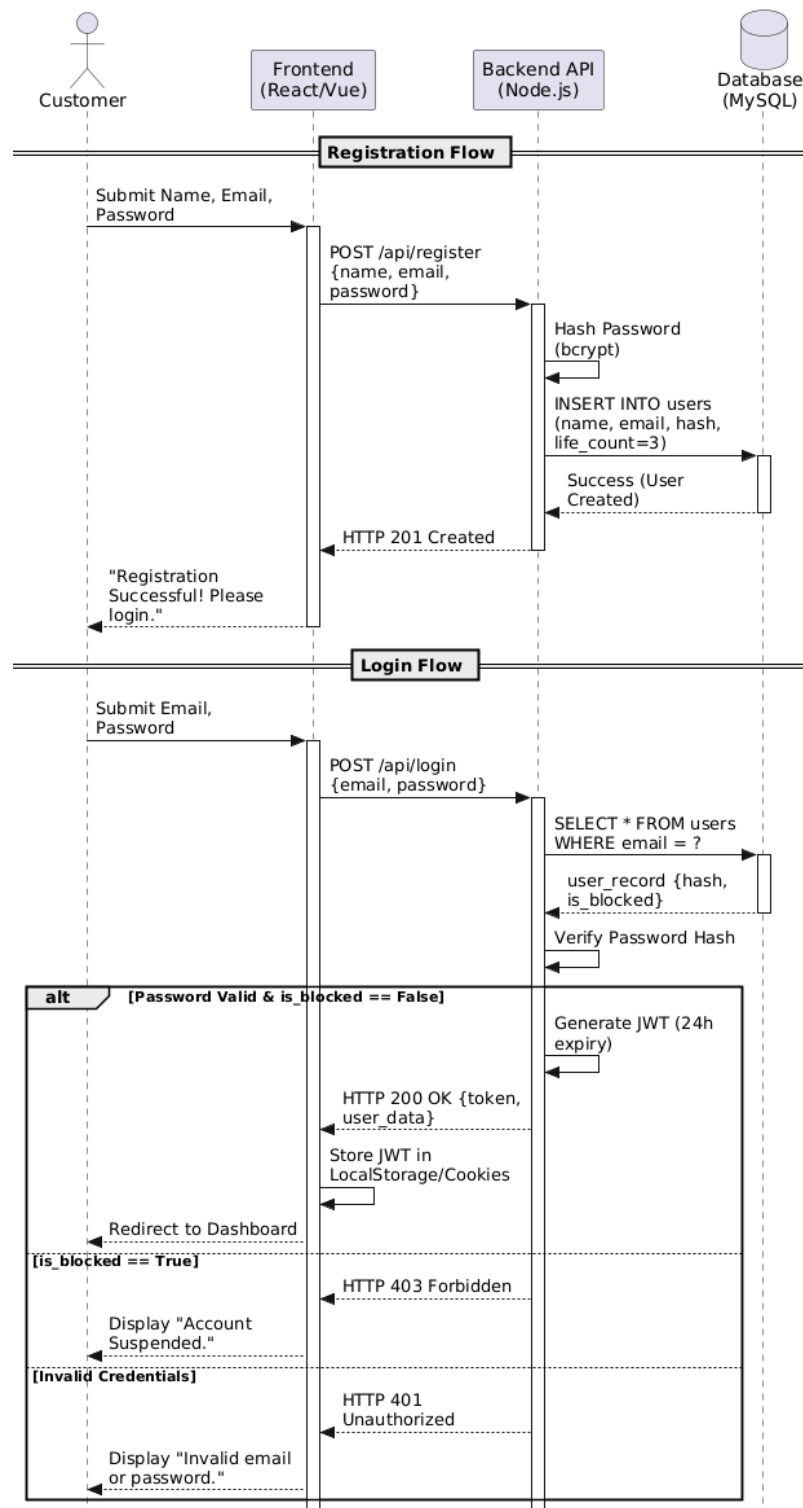


7.3 BarberHub ERD

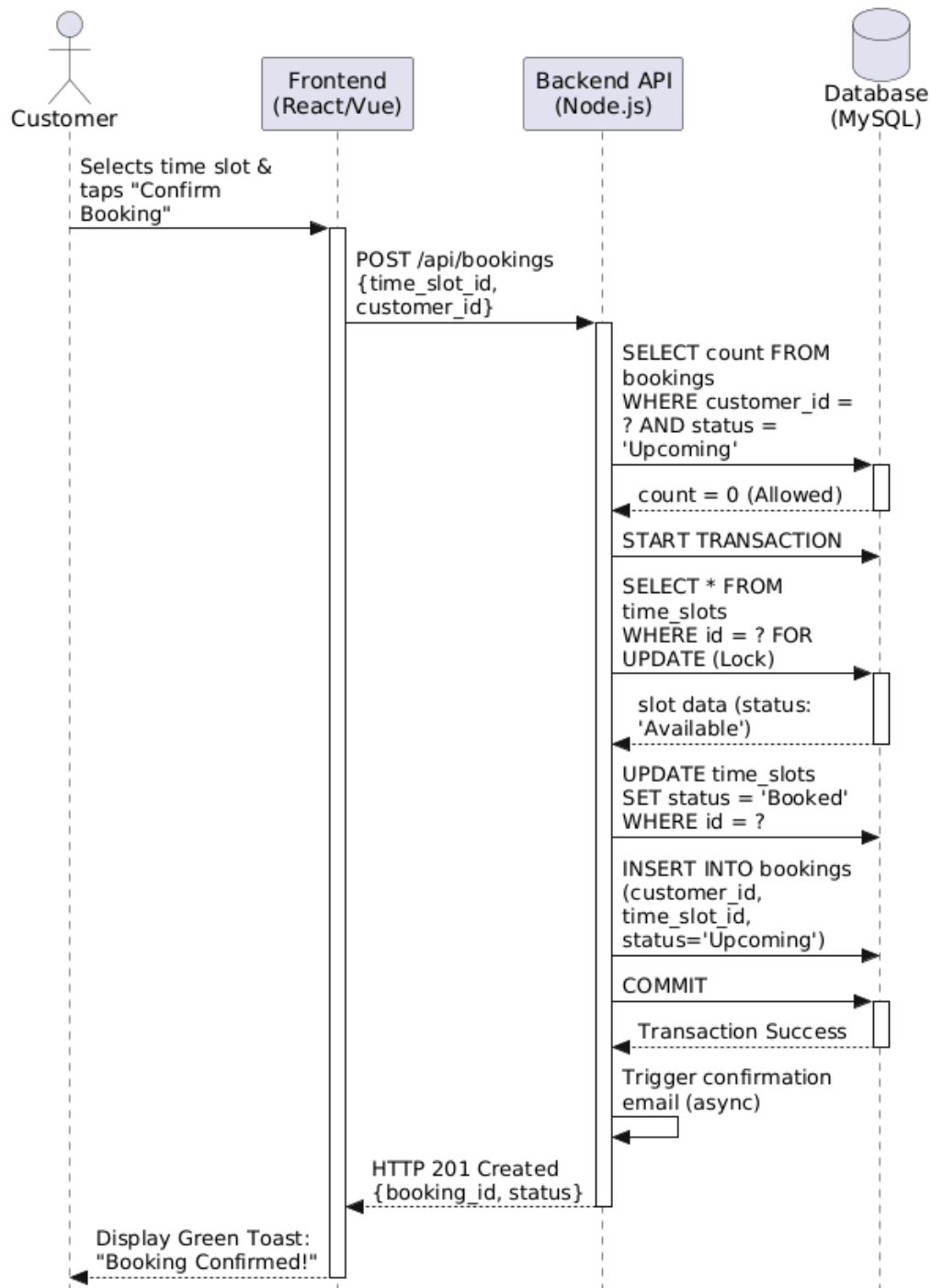


7.4 BarberHub Sequence Diagrams

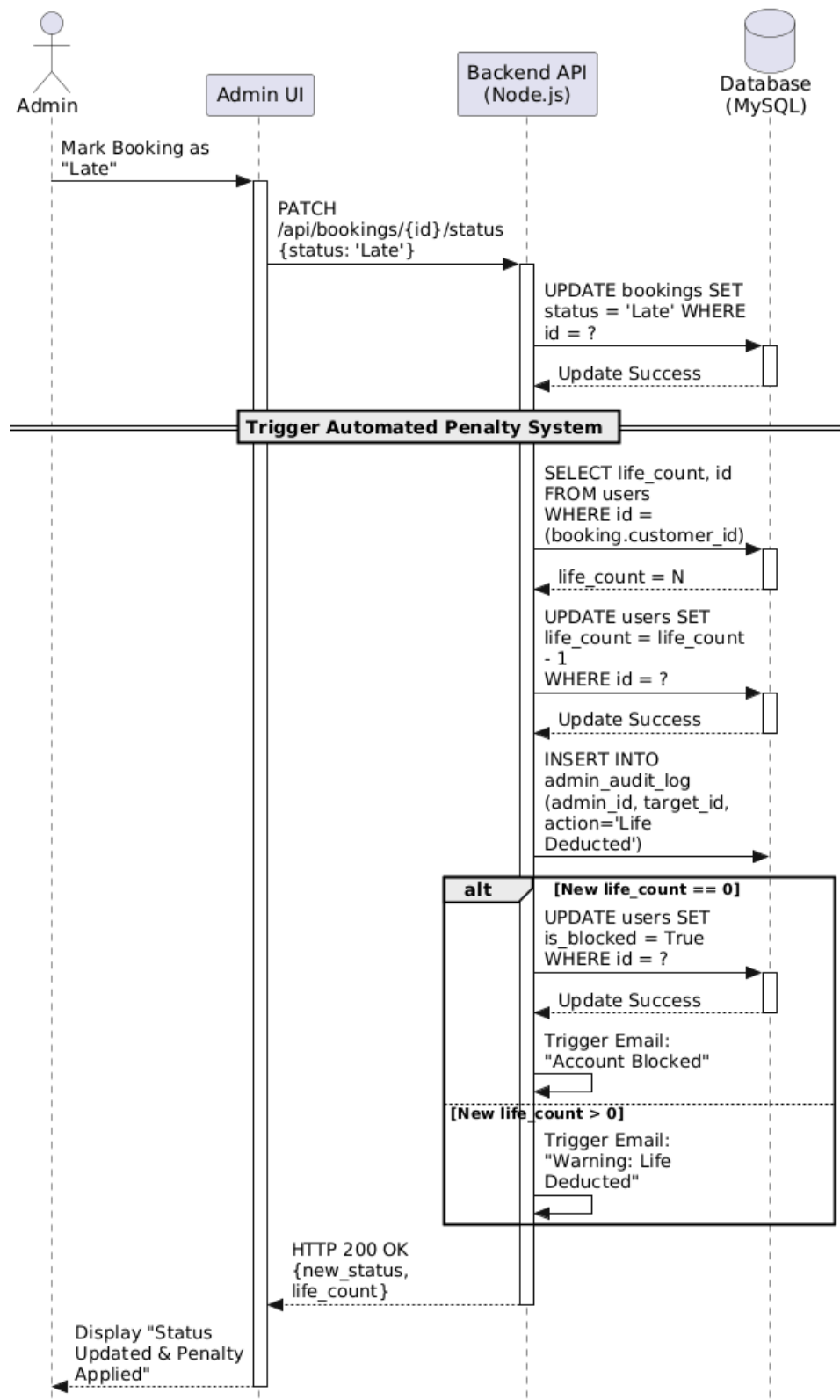
7.4.1 Registration & Login (Authentication)



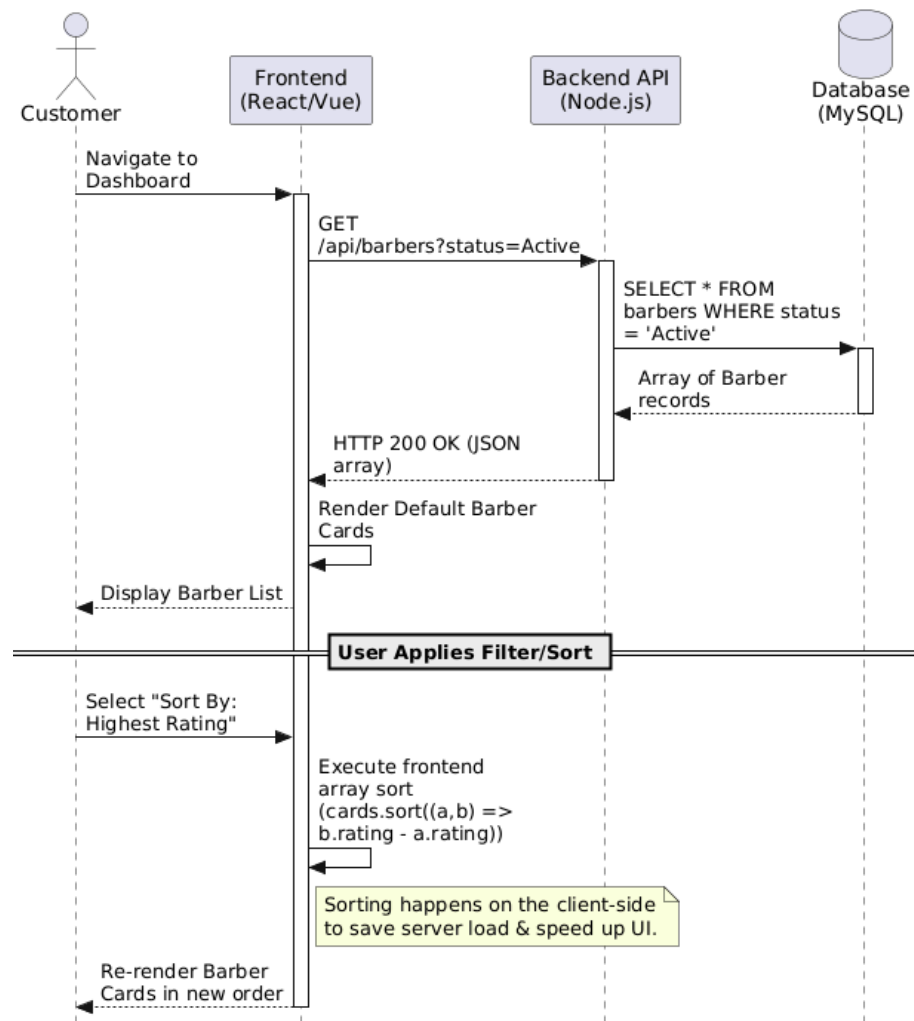
7.4.2 Sequence Diagram: Booking an Appointment



7.4.3 Sequence Diagram: Penalty Enforcement (3-Life System)



7.4.4 Sequence Diagram: Browse & Sort Barbers



8. Prototype

8.1 Link Towards Figma Link: [Group17 - BarberHub](#)